



Dynamic Pixel-to-Metre Calibration in Competitive Swimming Using Lane Rope Markers: A Computer Vision Proof of Concept

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INTRODUCTION

Traditional swimming analysis relies heavily on manual observation or expensive, intrusive sensor systems. While Computer Vision (CV) offers a non-intrusive alternative, tracking in aquatic environments introduces severe data degradation due to:

- Visual Noise: Light refraction, water surface turbulence, bubbles, and splashes.
- Environmental Variability: Inconsistent camera angles, distances, and lighting across different venues.

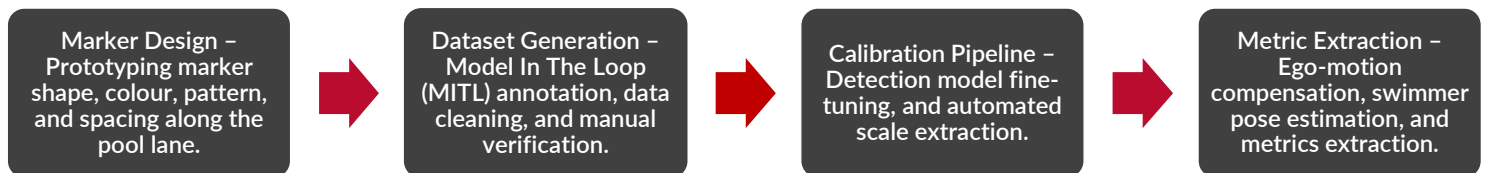
Because of these variations, extracting accurate metrics like speed is highly difficult without distinct calibration markers. This project addresses the lack of standardized, automated, and simple tracking systems. By focusing on minimal setup requirements, this system serves as a proof of concept to validate the feasibility and robustness of CV tracking in challenging aquatic environments.

AIM

The primary aim of this project is to develop a proof-of-concept system capable of automatically extracting swimming metrics from underwater video footage. For this purpose, the following objectives were defined:

- Marker Design: Design, prototype, and create a modular, underwater marker design that maximizes underwater visibility.
- Dataset Generation: Create and annotate a small dataset of swimming footage using the designed markers in the background.
- Calibration Pipeline: Establish a pipeline for dynamic pixel-to-metre scale extraction using the markers for calibration.
- Metric Extraction: Utilize the calibrated scale to provide useful kinematic metrics such as swimmer velocity and trajectory.

METHODOLOGY



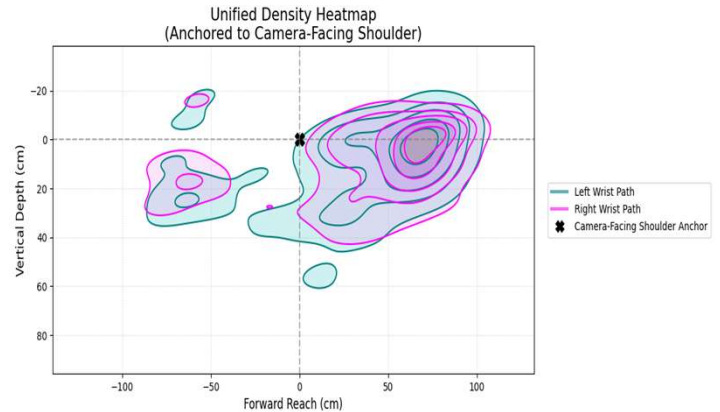
RESULTS

The system achieves a mean absolute split-time error of 1.005 s and a RMSE of 1.210 s across 18 checkpoints over a 50 m breaststroke sequence, demonstrating that marker-based ego-motion compensation and Kalman filtering can produce a usable positional trajectory from handheld, uncalibrated poolside footage. The primary positional limitation identified is the non-linear fluctuation in scale estimation on the outbound leg, resulting in alternating periods of overestimated and underestimated swimming speed rather than a strict, continuous monotonic drift. Distal keypoint tracking, particularly for the wrists, remains noisy in aquatic environments and represents the primary target for future refinement.

CONCLUSIONS AND FUTURE WORK

This project establishes a baseline for automated athletic tracking by extracting kinematic metrics from poolside video using composed open-source foundation models. To transition this proof of concept into a robust coaching tool, future work should focus on fine-tuning the pose estimator on aquatic-domain data to mitigate keypoint failures caused by occlusion. Additionally, transitioning from frame-by-frame filtering to temporal transformer architectures will improve tracking continuity during swimmer occlusions, while fusing monocular video with waterproof sensors will constrain linear scale drift.

Swimmer Wrist Symmetry Heatmap



Extracted vs. Ground Truth Trajectory

Phase	Distance (m)	True Time (s)	Auto Time (s)	Error (s)
Outbound	2.5	2.00	2.12	+0.12
Outbound	5.0	4.00	3.88	-0.12
Outbound	7.5	7.00	5.37	-1.63
Outbound	10.0	9.00	8.75	-0.25
Outbound	12.5	12.00	10.15	-1.85
Outbound	15.0	14.00	11.30	-2.70
Outbound	17.5	17.00	16.03	-0.97
Outbound	20.0	20.00	20.50	+0.50
Outbound	22.5	22.00	24.02	+2.02
Outbound	25.0	26.00	N/A	N/A
Return	22.5	27.00	28.13	+1.13
Return	20.0	29.00	29.90	+0.90
Return	17.5	32.00	31.37	-0.63
Return	15.0	35.00	33.52	-1.48
Return	12.5	38.00	37.17	-0.83
Return	10.0	41.00	40.07	-0.93
Return	7.5	43.00	42.20	-0.80
Return	5.0	46.00	45.42	-0.58
Return	2.5	49.00	48.37	-0.63
Return	0.0	51.00	N/A	N/A
Overall MAE				1.005 s
Overall RMSE				1.210 s